and shorter. When analyzed mathematically, the constant shortening of the cycles implies that they cannot stretch infinitely far into the past. Even in Tolman's cyclic framework, the universe would have a beginning.

Tolman's proposal invoked a spherical universe, which, as we've seen, has been ruled out by observations. But a radically new incarnation of cyclic cosmology, involving a flat universe, has recently been developed within string/M-theory. The idea comes from Paul Steinhardt and his collaborator Neil Turok of Cambridge University (with heavy use of results discovered in their collaborations with Burt Ovrut, Nathan Seiberg, and Justin Khoury) and proposes a new mechanism for driving cosmic evolution.[9] Briefly put, they suggest that we are living within a three-brane that violently collides every few trillion years with another nearby, parallel three-brane. And the "bang" from the collision initiates each new cosmological cycle.

The basic setup of the proposal is illustrated in Figure 13.7 and was suggested some years ago by Hořava and Witten in a noncosmological context. Hořava and Witten were trying to complete Witten's proposed unity among all five string theories and found that if one of the seven extra dimensions in M-theory had a very simple shape—not a circle, as in Figure 12.7, but a little segment of a straight line, as in Figure 13.7—and was bounded by so-called end-of-the-world branes attached like bookends, then a direct connection could be made between the Heterotic-E string theory and all others. The details of how they drew this connection are neither obvious nor of the essence (if you are interested, see, for example, *The Elegant Universe*, Chapter 12); what matters here is that it's a starting point that naturally emerges from the theory itself. Steinhardt and Turok enlisted it for their cosmological proposal.